

Cybersickness as Affected by Self-Motion Perception Instructions

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Abstract

Cybersickness is a common issue in virtual reality (VR), often caused by a mismatch between visual and physical motion cues. This study investigates whether altering users' self-motion perception can reduce cybersickness symptoms. Ten participants navigated a virtual environment using identical controls but receiving different movement perception instructions: either self-based (self-moving) or environment-based (world-moving). Cybersickness was assessed using mid-session 0-10 ratings, and pre/post-session CSQ-VR questionnaires. Post-session cybersickness scores were not shown to significantly differ between groups, but participants in the environment-based group reported significantly stronger agreement with the idea that the environment was moving around them. Contrary to the hypothesis, the environment-based group also experienced a slightly greater increase in cybersickness over time, though this could be attributed to the low sample size. These results suggest that movement perception can be influenced by instructions, though future research with larger sample sizes is needed to better understand the impact on cybersickness.

Introduction

When navigating in virtual reality (VR) environments, the brain is constantly being asked to complete a very tall task: perceive motion when one is not physically moving in any capacity. After all, a VR user is typically not physically moving at all, simply standing or sitting still, but may be experiencing many visual indications of movement within their virtual environment. This mismatch in virtual and physical stimuli can increase the difficulty of navigating within VR, and is also known to cause a type of motion sickness known as Visually Induced Motion Sickness (VIMS), or cybersickness (Keshavars et al. 2014). This experiment intends to explore an alternative method to avoid the issue of mismatched movement in the physical and virtual worlds, by altering the VR users' self-motion perception. If VR users can be convinced that they are stationary, and instead the virtual environment is what is moving around them, it is possible that they may experience a different intensity of cybersickness.

Background

Self-motion perception is influenced by various cues, including visual cues (e.g., displacement and optic flow), and physical cues (e.g., gravito-inertial forces and proprioception). Physical cues have been shown to be very important for the perception of self-motion. As shown in a 2002 study, when presented with solely visual cues of movement, participants consistently misjudged the distance of their movement, overestimating the distance by nearly three times. Receiving only physical motion cues, on the other hand, resulted in notably greater accuracy than with only visual cues, and combining both visual and physical cues resulted in the greatest accuracy of movement distance estimation (Harris et al. 2002). Having conflicting visual and physical cues is also generally seen as one of the primary causes of cybersickness. While there are still multiple competing theories as to the primary cause of cybersickness, what is generally

considered to be the leading theory is that of sensory conflict, stating that when multiple sensory organs give conflicting motion information, this causes a conflict in the brain that results in cybersickness (Kemeny et al. 2020). These findings suggest that the best way to combat cybersickness would be to find some way to make both the physical and visual cues the same, either by adding physical cues of movement, or by removing the visual cues of movement.

In order to incorporate physical cues in a VR application, external systems such as motion platforms or haptic feedback devices are often required. These external systems will typically result in the VR experience having a higher cost, larger footprint, and less mobility. As such, many VR applications still rely entirely on visual cues for self-motion perception. Without a cheap and practical method of creating physical self-motion cues, the alternative option is to remove the visual self-motion cues. In a typical VR application, the user will be sitting or standing in a static position, thus experiencing physical motion cues that indicate the lack of movement. As such, many VR applications will either take place entirely in one location, with no actual movement, or will use non-continuous movement such as teleportation, which allows VR users to instantly change position in the virtual environment without experiencing visual cues such as optic flow. Teleportation has consistently been found to be one of the best locomotion methods for reducing VIMS symptoms, outperforming continuous movement systems such as gaze-directed and pointing-based movement systems (Christou & Aristidou, 2017).

However, some applications require that users have the ability to continuously navigate an environment. In order to account for these situations without significantly increasing cybersickness, VR applications should allow users to change their positions in the environment without actually feeling like they are themselves moving through the environment.

If only provided with visual motion cues, there are two ways that these cues could be interpreted. Typically, VR users have self-based movement perception, where they perceive themselves to be moving around the environment, resulting in a mismatch between their visual motion perception and physical motion perception. However, it's also possible that this movement could be interpreted as the virtual environment moving around them while they themselves are completely stationary, or environment-based movement perception. If VR users could be consistently convinced to perceive motion in this way, this would result in their virtual motion perception matching their physical motion, possibly having an effect on the intensity of cybersickness, similar to in applications with static motion.

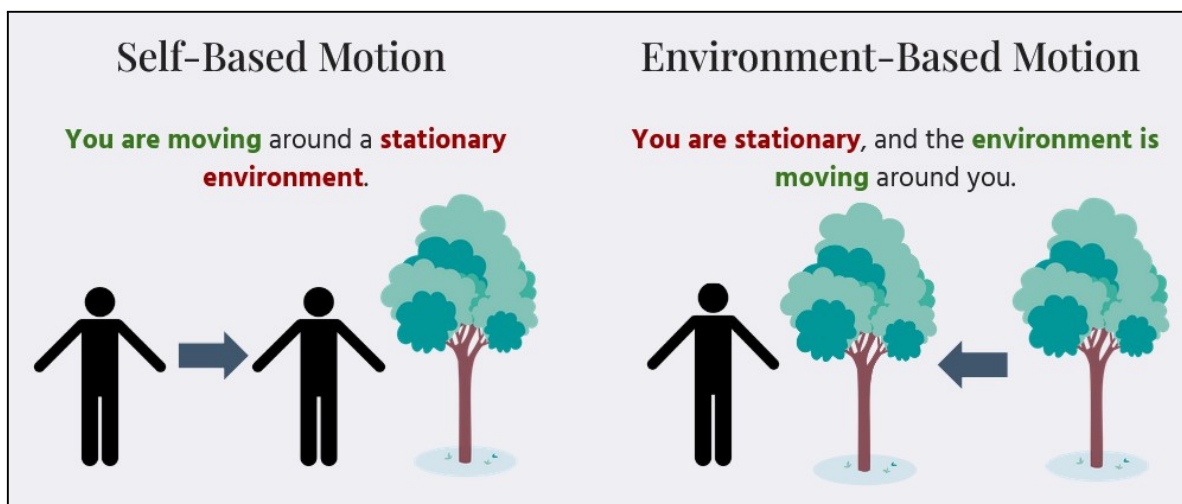


Figure 1: Self-Based Motion vs Environment-Based Motion

Simulation

A simple virtual environment was created within Unity, depicting an urban city block with multiple labeled stores and buildings. Twenty collectible spheres were spread throughout this environment. Users were able to navigate through this environment using the joystick and triggers on the Meta Quest 3 controller, collecting a sphere every time the sphere collides with their camera position, at which point the sphere disappears, and the user hears a voice stating the

number of spheres they have collected so far. Once the user has collected all 20 spheres, the simulation ends.



Figure 2: The virtual environment from above.

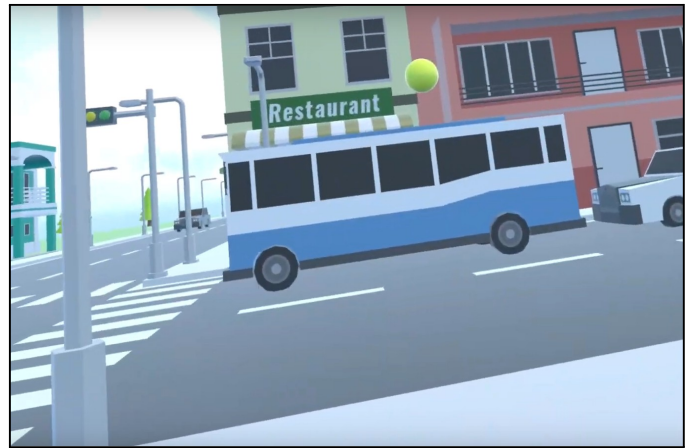


Figure 3: User perspective within simulation, with collectible yellow sphere.

Controls were set up such that tilting the joystick moved the camera position horizontally in the opposite direction of the tilt. For vertical movement, pressing the upper trigger moved the camera position directly downwards, while the lower trigger moved the camera position directly upwards. This control system was set up in a custom locomotion system script within Unity, with the forward source set to the controller's transform, and move speed set to five. The three movement input vectors (joystick and the two triggers), were added together to create one Vector3 indicating the direction and magnitude of movement. The magnitude of this Vector3 was clamped to one before being multiplied by the move speed, to maintain a consistent max speed of five while also allowing slower speeds if joysticks or triggers were pressed with less pressure. To change direction of the camera, users were told they could physically pivot in place.

This control scheme was chosen so that multiple types of movement perception could be described intuitively, either describing the controls as moving the camera in the opposite direction that the user inputs, or as moving the environment in the same direction as the user

inputs. While the underlying code of the simulation uses camera movement, this ultimately appears identical to environmental movement from the user's perspective, making both modes of movement perception completely valid ways to interpret the movement in the simulation.

Procedure

Ten healthy adults participated in the study. Participants were first asked to complete the Visually Induced Motion Sickness Susceptibility Questionnaire (VIMSSQ) to assess their individual susceptibility to cybersickness, followed by the Cybersickness in Virtual Reality Questionnaire (CSQ-VR) to assess any cybersickness symptoms that may have already been present at the beginning of the session (Keshavars et al. 2014; Kourtesis et al. 2023). Participants were also asked about their previous experience with virtual reality and first person video games with the following options for both: never used before, 1-5 times/year, 1-2 times/month, once/week, and multiple times/week. After completion of both of these surveys, participants donned the Meta Quest 3 headset, and were instructed to spend up to 20 minutes collecting spheres in the virtual environment. Half of the participants were given instructions explaining the control scheme using self-based (self-motion) movement perception, while the other half were given instructions explaining the control scheme using environment-based (world-motion) movement perception. The instruction scripts are given below.

Environment-Based Group: “In this simulation, you cannot move at all. While you are completely stationary, the environment around you is not, and you have the ability to control it. Using the joystick on your controller, you can make the entire world move in the direction you tilt. For example, if you tilt the joystick to the right, the world will move to the right, while you will stay still. To move the world vertically, you can press the upper trigger to move the world

upwards, and the lower trigger to move the world downwards. To look in different directions, you can physically pivot in place. Does this make sense?”

Self-Based Group: “In this simulation, you can move throughout the environment using the joystick and triggers on your controller. When you tilt the joystick in a certain direction, you will move in the opposite direction. For example, if you tilt the joystick to the right, you will move to the left. For vertical movement, you can press the upper trigger to move downwards, and the lower trigger to move upwards. To look in different directions, you can physically pivot in place. Does this make sense?”

While navigating the virtual environment, a beeping noise would play every 60 seconds, upon which participants were asked to rate their current intensity of cybersickness on a scale from zero to ten. A rating of zero indicated no cybersickness symptoms whatsoever, while a rating of ten indicated that they were feeling so sick that they would like to stop, at which point the session would end early. After the participants spent 20 minutes in the simulation, or collected all 20 spheres, they removed the headset and completed another CSQ-VR survey to assess their intensity of specific cybersickness symptoms during the simulation. They were then asked to rate the intuitiveness of the controls, the enjoyability of the simulation, and how much they agreed with the statements “I felt like I was moving around the environment” and “I felt like the environment was moving around me,” all on one to seven Likert scales. Finally, users were given a spatial awareness test, in which they were given a top down image of the virtual environment, and were asked to identify various buildings. The spatial awareness test is shown below.



Figure 4: Spatial Awareness Test

Results

Nine outcome variables were tested, including the total post-session CSQ-VR score, and its subscores for nausea, vestibular, and oculomotor symptoms, as well as participants' ratings of control intuitiveness, session enjoyability, and movement perception. Participants were also scored based on how they performed on the spatial awareness test. The results for each variable are presented below, along with the t-test results comparing the self-based and environmental-based groups.

	Mean (Self)	Mean (Env)	T	DF	P-value	95% Confidence Interval
CSQ-VR (Nausea)	4.8	6.4	-1.2649	7.7623	0.2426	(-4.5325, 1.3325)
CSQ-VR (Vestibular)	5.8	7.6	-1.3059	7.9288	0.2282	(-4.9836, 1.3836)
CSQ-VR (Oculomotor)	4.4	4.4	0	8	1	(-3.3576, 3.3576)
CSQ-VR (Total)	15.0	18.4	-1.027	7.9914	0.3345	(-11.0357, 4.2357)
Intuitiveness	3.2	3.2	0	7.0717	1	(-2.7721, 2.7721)
Enjoyability	4.6	4.0	0.5828	7.546	0.577	(-1.7993, 2.9993)
Self-Based Movement Perception	6.4	5.2	2.1909	5.8824	0.0719	(-0.1468, 2.5468)
Environment- Based Movement Perception	2.4	5.4	-2.357	7.9696	0.0463*	(-5.937, -0.063)
Spatial Awareness Test Score	1.4	1.2	-1.3927	5.3691	0.7638	(-1.3927, 1.7927)

In addition, multiple control variables were taken to account for possible variance in the post-session CSQ-VR score not related to group assignment. These included the VIMSSQ survey, a pre-session CSQ-VR survey, and 2 questions about participant's previous experience with VR and first-person videogames. Each of these control variables, and the means for the control and environment-based groups are shown in the table below.

Control Variable	Mean (Con)	Mean (Exp)
VIMSSQ	2.0	3.6
CSQ-VR (Nausea)	2.0	2.0
CSQ-VR (Vestibular)	2.0	2.0
CSQ-VR (Oculomotor)	2.4	3.4
CSQ-VR (Total)	6.4	7.4
Prior Experience (FPS)	0.8	2.4
Prior Experience (VR)	0.4	0.4

Two models were fit to predict the post-session CSQ-VR score using both the group assignment and control variables as predictors. The first model uses the group assignment, both pre-session surveys, and the prior experience questions as predictors of the post-session CSQ-VR score, while the second model also uses interaction terms between the group and surveys, but does not feature the prior experience questions as predictors. The coefficient estimates for both models are shown below.

Model 1	Estimate	Std. Error	T-value	P-value
Intercept	14.1444	19.1684	0.738	0.502
Environment Based	1.8394	5.8694	0.313	0.770
CSQ-VR	-0.0456	2.9368	-0.016	0.988
VIMSSQ	0.3496	1.4862	0.234	0.826
Prior Experience (FPS)	0.6542	3.0229	0.216	0.839
Prior Experience (VR)	-0.1878	5.0235	-0.037	0.972

Model 2	Estimate	Std. Error	T-value	P-value
Intercept	35.2290	21.6833	1.625	0.180
Environment Based	-46.3147	40.0971	-1.155	0.312
CSQ-VR	-3.1489	3.2892	-0.957	0.393
VIMSSQ	-0.0382	1.0091	-0.038	0.972
Env:CSQ-VR	7.0917	6.2007	1.144	0.317
Env:VIMSSQ	0.1239	2.1976	0.056	0.958

Mid session cybersickness ratings were also taken once every 60 seconds, each on a scale of 0-10. Each participant's ratings over time are shown in the figure below.

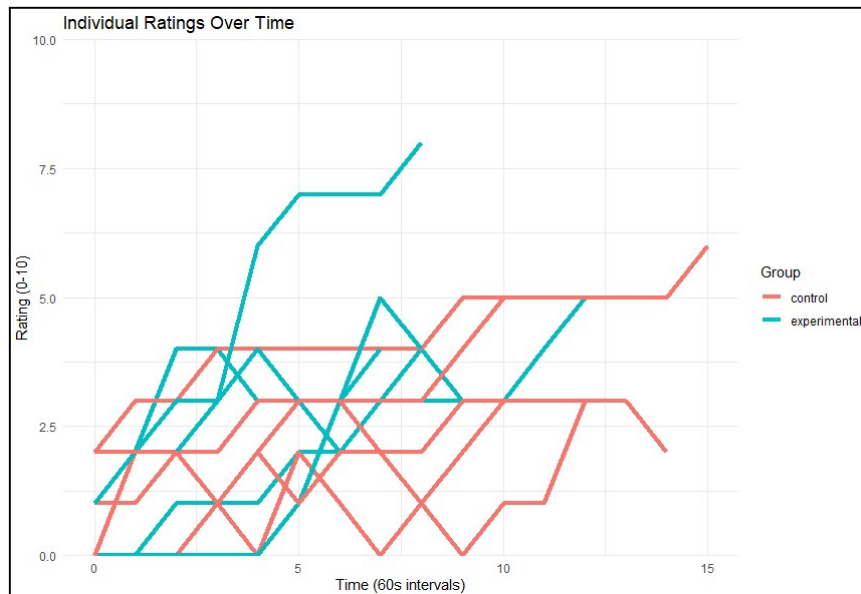


Figure 5: Individual Cybersickness Ratings Over Time

These ratings were analyzed with a linear mixed-effects model, which predicts cybersickness ratings depending on the participant's group assignment and the time point at which the rating is given. The coefficient estimates for this model are shown below.

	Estimate	Std. Error	DF	T-value	P-value
Intercept	1.1940	0.6259	9.1072	1.907	0.0884
Time	0.2042	0.0264	110.323	7.921	2.03e-12***
Environment Based	-0.1904	0.8877	9.2097	-0.214	0.8348
Time:Env	0.1757	0.0434	111.0353	4.046	9.65e-05***

Discussion

Given the small sample size of 10 participants, the results of the post-session CSQ-VR scores were not statistically significant, meaning there was insufficient evidence to reject the null hypothesis that group assignment had no effect on the post-session CSQ-VR score. Through power analysis using G*Power, we can see that 44 participants would be required for a large effect size, 92 participants for a medium effect size, and 642 participants for a small effect size. Unfortunately, due to time constraints, participant counts this high was unable to be reached. However, the study procedure could be extended in the future with larger sample sizes to enhance the statistical power.

Similarly, participants' ratings of control intuitiveness, session enjoyability, and their spatial awareness test scores were not shown to be significantly different between the two groups. This likely indicates that there is not a large difference between these scores depending on the perception instructions provided to participants. However, with only 10 participants, the possibility of small or medium effect sizes cannot be ruled out.

While the post-session CSQ-VR scores can't be significantly analyzed, that doesn't mean there was a complete lack of statistically significant results. For one, participants in the environment-based group were shown to give significantly higher agreement ratings for the statement "I felt like the environment was moving around me" compared to the self-based group.

This suggests that the instructions influenced how participants perceived movement within the virtual environment, with those in the environment-based group being more likely to perceive themselves as staying stationary while the environment moved around them.

However, this effect does not also apply to the counterstatement, “I felt like I was moving around the environment.” While the self-based group did rate higher levels of agreement with this statement than the environment-based group did, the difference between the means was much smaller and less statistically significant. Reading participant comments provides some insight into this result, with one participant in the environment-based group noting, “I would think ‘I want to go over there’ before moving the world.” This comments highlights that, even with the alternate instructions, participants in the environment-based group still often conceptualizing themselves as changing position within the world, albeit with a different perception of how motion occurred.

Additionally, mid-session cybersickness ratings, which were taken at 60 second intervals throughout the session, revealed an increase in cybersickness ratings over time. On average, participants’ cybersickness ratings increased by 0.2042 for each additional minute spent in the session. The interaction between time and group assignment was also found to have a highly significant result, suggesting that those in the environment-based group experienced a slightly greater increase in cybersickness over time compared to the self-based group. This contradicts the initial hypothesis, as those in the environment-based group experienced a faster increase in cybersickness. However, due to the small sample size, individual variance between participants cannot be ruled out as a factor.

Conclusion

Although the small sample size limited the ability to draw definitive conclusions, several interesting insights were still uncovered. Notably, the perception instructions were shown to affect participants' perception of movement within VR, with those in the environment-based group more likely to report that the environment was moving around them. However, this was not shown to decrease cybersickness levels as initially hypothesized, with mid-session cybersickness ratings increasing faster among participants in the environment-based group, though this may have resulted due to a low sample size. These findings indicate that there is very likely a difference in the way these instructions affect the perception of movement in VR, and as such future research should look into repeating similar procedure with larger participant pools to more definitively assess potential effects perception instructions could have on VR experiences and cybersickness.

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